

2. Gauss-Elimination Method:

To understand gauss-elimination method let's take an example of 3x3 system.

$$10x_1 + 3x_2 + 1x_3 = 19$$

$$3x_1 + 10x_2 + 2x_3 = 29$$

$$1x_1 + 2x_2 + 10x_3 = 35$$

$$A = \begin{bmatrix} 10 & 3 & 1 \\ 3 & 10 & 2 \\ 1 & 2 & 10 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, B = \begin{bmatrix} 19 \\ 29 \\ 35 \end{bmatrix}$$

Step 1: Write the augmented matrix $A_B = [A:B]$

$$A_b = \begin{bmatrix} 10 & 3 & 1 & 19 \\ 3 & 10 & 2 & 29 \\ 1 & 2 & 10 & 35 \end{bmatrix}$$

Step 2: Make all the entries zero below each pivot entry.

$$A_b = \begin{bmatrix} \color{red}{10} & 3 & 1 & 19 \\ 3 & 10 & 2 & 29 \\ 1 & 2 & 10 & 35 \end{bmatrix} ; \quad m = \frac{3}{10} = 0.3 \Rightarrow R_2 - mR_1$$

$$A_b = \begin{bmatrix} \color{red}{10} & 3 & 1 & 19 \\ 0 & 9.1 & 1.7 & 23.3 \\ 1 & 2 & 10 & 35 \end{bmatrix} ; \quad m = \frac{1}{10} = 0.1 \Rightarrow R_3 - mR_1$$

$$A_b = \begin{bmatrix} \color{red}{10} & 3 & 1 & 19 \\ 0 & 9.1 & 1.7 & 23.3 \\ 0 & 1.7 & 9.9 & 33.1 \end{bmatrix}$$

$$A_b = \begin{bmatrix} 10 & 3 & 1 & 19 \\ 0 & 9.1 & 1.7 & 23.3 \\ 0 & 1.7 & 9.9 & 33.1 \end{bmatrix} ; \quad m = \frac{1.7}{9.1} = 0.1868 \Rightarrow R_3 - mR_2$$

$$A_b = \begin{bmatrix} 10 & 3 & 1 & 19.0000 \\ 0 & 9.1 & 1.7 & 23.3000 \\ 0 & 0 & 9.5824 & 28.7473 \end{bmatrix}$$

Step 3: Perform Back substitution to get solution.

$$10x_1 + 3x_2 + 1.0x_3 = 19.0000$$

$$9.1x_2 + 1.7x_3 = 23.3000$$

$$9.5824x_3 = 28.7473$$

$$\Rightarrow x_3 = \frac{28.7473}{9.5824} = 3.0000$$

$$\Rightarrow x_2 = \frac{23.3000 - 1.7(3)}{9.1} = 2.0000$$

$$\Rightarrow x_1 = \frac{19.0000 - 1(3) - 3(2)}{10} = 1.0000$$

MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

Find Files Find Go To Comment Indent Breakpoints Run Run and Advance Run Section Advance Run and Time

FILE NAVIGATE EDIT BREAKPOINTS RUN

E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab

Editor - E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab\GaussElimination_method.m

Simpson_38Rule.m x fixed_point.m x GaussElimination_method.m x Simpson_13Rule.m x +

```
1  %%% Gauss Elimination Method Matlab CODE
2  %%%By Zamir Hussain
3  A=input('Enter your coeddicient matrix:');
4  b=input('Enter the source vector:');
5  N=length(b);
6  x=zeros(N,1);
7  Aug=[A b]
8  for j=1:N-1
9      for i=j+1:N
10         m=Aug(i,j)/Aug(j,j);
11         Aug(i,:)=Aug(i,:)-m*Aug(j,:);
12     end
13 end
14 Aug
15 %%%%%%%%%End of the code%%%%%%%%%
```





MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

Find Files Compare Print Go To Find Insert Comment Indent Breakpoints Run Run and Advance Run and Time

FILE NAVIGATE EDIT BREAKPOINTS RUN

E: > University of wah > Fall Semester 2022 > Lecture Notes Numerical Analysis > Matlab Lab

Command Window

```
19
29
35

Aug =

    10     3     1    19
     3    10     2    29
     1     2    10    35

Aug =

    10.0000     3.0000     1.0000    19.0000
         0     9.1000     1.7000    23.3000
         0         0     9.5824    28.7473

GaussElimination_method
fx Enter your coeddicient matrix:
```



MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

Find Files Find Compare Print Go To Find Insert Comment Indent Breakpoints Run Run and Advance Run Section Advance Run and Time

E: > University of wah > Fall Semester 2022 > Lecture Notes Numerical Analysis > Matlab Lab >

Command Window

```

0      9.1000      1.7000      23.3000
0      0      9.5824      28.7473

GaussElimination_method
Enter your coedddicient matrix:randi([1,10],10,10)

A =

     9     2     7     8     5     3     8     9     4     1
    10    10     1     1     4     7     3     3     9     1
     2    10     9     3     8     7     6     9     6     6
    10     5    10     1     8     2     7     3     6     8
     7     9     7     1     2     2     9    10    10    10
     1     2     8     9     5     5    10     4     3     2
     3     5     8     7     5    10     6     2     8     6
     6    10     4     4     7     4     2     3     8     5
    10     8     7    10     8     6     2     7     4     1
    10    10     2     1     8     3     3     5     6     4

fx Enter the source vector:

```

MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

Find Files Find Compare Print Go To Find Insert Comment Indent Breakpoints Run Run and Advance Run and Time

E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab

Command Window

```

3      5      8      7      5      10      6      2      8      6      3
6      10     4      4      7      4      2      3      8      5      7
10     8      7     10     8      6      2      7      4      1      7
10     10     2      1      8      3      3      5      6      4      8

Aug =

    9.0000    2.0000    7.0000    8.0000    5.0000    3.0000    8.0000    9.0000    4.0000    1.0000    2.0000
         0    7.7778   -6.7778   -7.8889   -1.5556    3.6667   -5.8889   -7.0000    4.5556   -0.1111    5.7778
         0         0   15.7714   10.9143    8.8000    1.8286   11.4571   15.6000   -0.4857    5.9143   -3.5429
         0         0         0   -8.2844    0.4094   -3.1812   -3.1585   -9.0924    0.0716    5.1875    2.7572
         0         0         0         0   -5.0477   -3.5323    3.8058    5.2976    2.7483    4.2854   -4.3564
         0         0         0         0         0    1.2684    2.6917   -8.1947    1.8698    1.1135    8.6243
         0         0         0         0         0         0   -13.8850   26.9172   -4.4751   -2.2507  -35.9066
         0         0         0         0         0         0    -18.1175    5.9353    7.9667   10.8454
         0         0         0         0   -0.0000   -0.0000         0         0   -6.3361   -5.7952    6.9612
         0         0    0.0000         0   -0.0000   -0.0000    0.0000         0         0    4.4826   -1.7661

fx >> |

```

MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

Find Files Compare Print Go To Find Insert Comment Indent Breakpoints Run Run and Advance Run Section Run and Time

Current Folder

extra

PDF MATLAB CODE

All non linear equation matlab code...

BC Method.m

Bisection mehod code 2.m

Bisection_code.m

Bisection_method.m

Cramer_Rule.m

fixed_point.m

Select a file to view details

Workspace

Name	Value
A	10x10 double
Aug	10x11 double
b	[2;8;4;6;2;7;...

Editor - E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab\GaussElimination_method.m*

```

3 - A=input('Enter your coeddicient matrix:')
4 - b=input('Enter the source vector:')
5 - N=length(b);
6 - x=zeros(N,1);
7 - Aug=[A b]
8 - for j=1:N-1
9 -     for i=j+1:N
10 -         m=Aug(i,j)/Aug(j,j);
11 -         Aug(i,:)=Aug(i,:)-m*Aug(j,:);
12 -     end
13 - end
14 - Aug
15 - X(N)=Aug(N,N+1)/Aug(N,N);
16 - for k=N-1:-1:1
17 -     x(k)=Aug(k,N+1)-Aug(k,k+1:N)*x(k+1:N);
18 - end
19 - %%%%%%%%%End of the code%%%%%%%%

```

Command Window



MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

Find Files Compare Go To Find Insert Comment Indent Breakpoints Run Run and Advance Run and Time

Search Documentation

E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab

Command Window

0	0.0000	-0.0000	0.0000	0	0	0	1.6588	0.7841	-2.8194	-3.1913
0	0.0000	-0.0000	0.0000	0	0	0	0	2.7363	0.2752	-3.4619
0	0.0000	-0.0000	0.0000	0.0000	0	0	0	0	-15.1099	-10.2863

>> Aug=[A,b]

Aug =

3	1	3	5	3	4	2	3	5	4	3
1	5	5	4	1	2	2	2	4	2	4
2	1	1	2	5	2	5	4	3	5	5
5	4	2	3	5	3	1	1	3	3	3
1	5	1	3	3	1	1	4	3	2	9
5	5	1	1	3	1	1	1	2	5	2
3	1	5	2	2	5	4	2	3	5	3
5	2	3	1	5	5	4	4	3	3	2
1	2	3	1	2	3	4	4	5	4	3
3	5	1	2	1	1	3	1	4	3	5

fx >>

MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

Find Files Compare Go To Find Insert Comment Indent Breakpoints Run Run and Advance Run and Time

FILE NAVIGATE EDIT BREAKPOINTS RUN

E: > University of wah > Fall Semester 2022 > Lecture Notes Numerical Analysis > Matlab Lab >

Command Window

```

5 2 3 1 5 5 4 4 3 3 2
1 2 3 1 2 3 4 4 5 4 3
3 5 1 2 1 1 3 1 4 3 5

>> rref(Aug)

ans =

1.0000    0    0    0    0    0    0    0    0    0    -2.1262
0    1.0000    0    0    0    0    0    0    0    0    2.0050
0    0    1.0000    0    0    0    0    0    0    0    -2.2736
0    0    0    1.0000    0    0    0    0    0    0    1.1401
0    0    0    0    1.0000    0    0    0    0    0    -0.1751
0    0    0    0    0    1.0000    0    0    0    0    3.0218
0    0    0    0    0    0    1.0000    0    0    0    0.6430
0    0    0    0    0    0    0    1.0000    0    0    -0.1364
0    0    0    0    0    0    0    0    1.0000    0    -1.3336
0    0    0    0    0    0    0    0    0    1.0000    0.6808

```

fx >>



MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

Find Files Compare Print Go To Find Insert Comment Indent Breakpoints Run Run and Advance Run Section Advance Run and Time

Search Documentation

E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab

Current Folder

- extra
- PDF MATLAB CODE
 - All non linear equation matlab code...
 - BC Method.m
 - Bisection mehod code 2.m
 - Bisection_code.m
 - Bisection_method.m
 - Cramer_Rule.m
 - fixed_point.m

Details

Select a file to view details

Workspace

Name	Value
A	10x10 double
ans	10x11 double
Aug	10x11 double
b	[3;4;5;3;9;2;...

Editor - E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab\GaussElimination_method.m

Simpson_38Rule.m fixed_point.m GaussElimination_method.m Simpson_13Rule.m

```

1  %% Gauss Elimination Method Matlab CODE
2  %% By Zamir Hussain
3  A=input('Enter your coeddicient matrix:');
4  b=input('Enter the source vector:');
5  N=length(b);
6  x=zeros(N,1);
7  Aug=[A b]
8  for j=1:N-1
9      for i=j+1:N
10         m=Aug(i,j)/Aug(j,j);
11         Aug(i,:)=Aug(i,:)-m*Aug(j,:);
12     end
13 end
14 Aug
15 X(N)=Aug(N,N+1)/Aug(N,N);
16 for k=N-1:-1:1
17     x(k)=Aug(k,N+1)-Aug(k,k+1:N)*x(k+1:N);

```

Command Window

MATLAB R2015a

HOME PLOTS APPS EDITOR PUBLISH VIEW

File Edit Breakpoints Run

Current Folder: E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab

Editor: E:\University of wah\Fall Semester 2022\Lecture Notes Numerical Analysis\Matlab Lab\GaussElimination_method.m

```

3 - A=input('Enter your coeddicient matrix:')
4 - b=input('Enter the source vector:')
5 - N=length(b);
6 - x=zeros(N,1);
7 - Aug=[A b]
8 -     for j=1:N-1
9 -         for i=j+1:N
10 -             m=Aug(i,j)/Aug(j,j);
11 -             Aug(i,:)=Aug(i,:)-m*Aug(j,:);
12 -         end
13 -     end
14 -     Aug
15 -     X(N)=Aug(N,N+1)/Aug(N,N);
16 -     for k=N-1:-1:1
17 -         x(k)=Aug(k,N+1)-Aug(k,k+1:N)*x(k+1:N);
18 -     end
19 -     %%%%%%%%%End of the code%%%%%%%%

```

Workspace:

Name	Value
A	10x10 double
ans	10x11 double
Aug	10x11 double

Thank You!!!

